

Executive Report

Income Prediction Model – Final Deployment Report

1. Business Goal

The objective of this project was to develop a machine learning model capable of predicting whether an individual's annual income exceeds **\$50,000** using demographic and socioeconomic information from the Adult Census dataset. The model is intended to support targeted marketing and premium service allocation by identifying individuals who are most likely to belong to the higher-income group.

Since incorrectly classifying low-income individuals as high-income customers can lead to unnecessary marketing costs and inefficient resource allocation, **Precision** was selected as the primary business evaluation metric. Maximizing precision ensures that individuals predicted as high-income are highly likely to belong to the target group while minimizing false positive predictions.

2. Data Used

The project used the **Adult Census Income Dataset** obtained from OpenML, containing approximately **48,842 records** with demographic, educational, occupational, and financial information. Features included age, education, workclass, occupation, marital status, relationship, race, sex, capital gain, capital loss, hours worked per week, and native country.

To ensure unbiased evaluation, the data was divided into **training, development (validation), and hold-out test** sets. The hold-out test set remained completely untouched until final model evaluation.

Several business-driven features were engineered to improve predictive performance, including:

- Age Bucket
- Hours Worked Bucket
- Capital Gain Flag
- Log Capital Gain
- Higher Education Flag
- Education × Hours Interaction (edu_hours)

Mutual Information analysis confirmed that engineered features added predictive value. The most informative engineered variables were:

Engineered Feature	Mutual Information
Log Capital Gain	0.0794
Education × Hours	0.0783
Higher Education Flag	0.0490
Capital Gain Flag	0.0322

3. Final Model & Deployment Artifact

Four supervised learning models were evaluated:

- Logistic Regression
- Random Forest
- HistGradientBoosting
- Stacking Ensemble

Among all models, the **Random Forest Classifier** achieved the highest Precision (**86.21%**) while maintaining strong predictive performance across all evaluation metrics and was therefore selected as the final deployment model.

The complete deployment pipeline includes:

- Automated Feature Engineering
- Missing Value Imputation
- Feature Scaling
- One-Hot Encoding
- Random Forest Classification

The trained pipeline was exported as:

adult_income_pipeline.pkl

An accompanying inference script validates inputs, handles unseen categorical values, predicts probabilities, applies the decision threshold, and supports production deployment.

4. Hold-Out Performance & Decision Threshold

The final model was evaluated on the untouched hold-out test set using multiple classification metrics.

Metric	Final Result
Accuracy	83.97%
Precision	86.21%

Recall	39.31%
F1 Score	54.00%
ROC-AUC	0.9014
PR-AUC	0.7717
Inference Time	0.2012 sec

The production system uses a **decision threshold of 0.50**, providing the highest precision while keeping inference fast enough for real-time prediction. Because Precision is the primary business objective, this threshold minimizes costly false positive predictions.

5. Model Interpretability

Model transparency was evaluated using **Random Forest Feature Importance**, **Permutation Importance**, and **SHAP (SHapley Additive Explanations)**.

The most influential variables were consistently:

Rank Important Feature

- 1 Education Number
- 2 Capital Gain
- 3 Education
- 4 Occupation
- 5 Native Country
- 6 Workclass
- 7 Hours Worked per Week
- 8 Capital Loss

SHAP local explanations were generated for representative **True Positive**, **False Positive**, and **False Negative** predictions. These visualizations demonstrated how individual features contributed toward each prediction, improving transparency and enabling business users to understand why the model reached a particular decision.

6. Fairness Evaluation

Model fairness was assessed by evaluating predictive performance across demographic groups including **Sex** and **Race**. Precision was compared separately for each protected group to identify potential disparities.

Although slight differences existed across demographic groups, no severe imbalance was observed during evaluation. Continuous monitoring is nevertheless recommended to ensure

fairness throughout deployment. Future improvements may include fairness-aware preprocessing, threshold adjustment, and periodic auditing of subgroup performance.

7. Deployment & Monitoring

The model is deployment-ready through the exported pipeline and supports both **CSV** and **dictionary** inputs. The inference pipeline performs automatic input validation, safely handles previously unseen categorical values using One-Hot Encoding, predicts class probabilities, and produces reliable predictions for production use.

A monitoring framework was also defined to continuously observe:

- Data drift
- Prediction distribution
- Model confidence
- Missing values
- Unseen categories
- Precision degradation
- Fairness metrics

Automatic retraining is recommended every **three months**, or sooner if monitoring indicates significant performance degradation.

8. Recommended Next Steps

A phased deployment strategy is recommended to reduce operational risk.

Phase 1 – Shadow Deployment: Run the model alongside the current business workflow without affecting operational decisions.

Phase 2 – A/B Testing: Deploy the model to approximately **20%** of incoming cases while the remaining **80%** continue using the existing process. Compare key business metrics including Precision, False Positive Rate, Customer Conversion Rate, Operational Cost, and Customer Response.

Phase 3 – Full Production Deployment: Following successful A/B testing, deploy the model to all production traffic while continuously monitoring model drift, prediction quality, fairness metrics, and business performance. Scheduled retraining every three months will help maintain long-term predictive accuracy and reliability.